

Shangyu Andrew Xu

☎ (203) 909-8168 | ✉ andrew.xu@yale.edu | in sandrewxu | 🌐 sandrewxu

EDUCATION

Yale University

Aug 2023 – May 2027 (Expected)

B.S./M.S. in Computer Science, B.S. in Mathematics (Intensive)

GPA: 3.98/4.0

Coursework: CS 477: Natural Language Processing (Grad), S&DS 365: Intermediate ML (Grad), CS 486: Probabilistic ML (Grad), CS 483: Deep Learning on Graphs (Grad), CS 323: Systems Programming, CS 365: Algorithms, CS 462: Spectral Graph Theory (Grad), MATH 226: Linear Algebra (Intensive)

Activities & Awards: Yale Glee Club, MathCounts Outreach, 5x AIME Qualifier

EXPERIENCE

Machine Learning Researcher

Aug 2024 – Present

Natural Language Processing (NLP) Lab, Yale University

New Haven, CT

- Research under Prof. Arman Cohan focuses on reasoning-intensive retrieval and geometric embeddings
- Adapting top-performing retrievers on the Massive Text Embedding Benchmark (MTEB) to increase performance reasoning-intensive Question-Answer benchmarks and generate higher-quality output
- Using hyperbolic distance as a loss function for contrastive learning; improved performance on BEIR benchmark by 10 points

Head Teaching Assistant

Aug 2024 – Present

CS223: Data Structures and Programming Techniques, Yale University

New Haven, CT

- Curate and teach weekly section material for 200 students per semester about programming in C
- Developed and created 350+ public and private test cases for 6 brand-new coding projects in Fall 2024

Machine Learning Engineer Intern

May 2024 – Aug 2024

ActualFood (Startup)

Remote

- Built a full-stack iOS application in Swift focused on neural network-powered gesture recognition
- Designed and trained a CNN in PyTorch (inspired by MobileNetV4) for real-time hand gesture classification
- Optimized inference time for on-device model deployment

Business Manager

May 2024 – Present

Doox of Yale A Cappella

New Haven, CT

- Organized 70+ performances earning \$50k+ in gross revenue and planned 40 days of domestic and international tour across 3 continents
- Recorded 19 singers in a studio and collaborated with industry professionals to edit, mix, and master tracks for a 14-song a cappella album
- Lead the initiative to incorporate Doox of Yale as a 501(c)(3) non-profit organization

PUBLICATIONS

Centralized Selection with Preferences in the Presence of Biases | *ICML 2024*

Nov 2023 – Jul 2024

- <https://arxiv.org/pdf/2409.04897>
- Co-first author and conference presenter; authored with Dr. Nisheeth Vishnoi, Dr. Elisa Celis, and Dr. Amit Kumar
- Researched the mathematical optimization of a centralized selection algorithm under unknown bias factors
- Spearheaded empirical section with simulations of real-world scenarios where assumptions did not hold

TECHNICAL SKILLS

Languages: Python, C/C++, Swift

Machine Learning Tools: PyTorch, TensorFlow, Hugging Face Transformers, Matplotlib, NumPy, Microsoft Azure

Developer Tools: Git, VS Code, Xcode, Microsoft Azure